

Introduction to Computer Programming, Spring Term 2023
Practice Assignment 1

Discussion: 25.02.23 - 02.03.23

Exercise 1-1 Errors and Conventions
To be discussed in tutorial

In the following lab exercises, follow the instructions carefully and observe the outcome. In each case you should be able to explain why the output is as it is.

For each of the .java classes in the “ Lab 1 Exercises.zip ” zipped folder, compile the files on JCreator. You are asked to define in each file the lines which have the following faults:

- a) Syntax Errors
- b) Runtime Errors
- c) Logic Errors

Furthermore, specifically define the errors. For example, the following line of code:

```
String Word = "Hello World!";  
system.out.print(Word)
```

has two Syntax Errors and 1 bad identifier naming. The correct solution is:

```
String word = "Hello World!";  
System.out.println(word);
```

Exercise 1-2 Java Expressions
To be discussed in tutorial

- a) `System.out. println (-( -1));`

Solution:

Syntactically correct, value 1.

- b) `System.out. println ( --1 );`

Solution:

Syntactically incorrect. “--” is an operator in Java that only works on variables. Correct uses of this operator (and the related “++” operator) are: “++i”, “--i”, “i++”, and “i--”.

- c) `System.out. println (2*4(-6));`

Solution:

Syntactically incorrect.

There is an operator missing between 4 and (-6) .

d) `System.out.println ((12)*(4)−6);`

Solution:

Syntactically correct, value 42.

e) `System.out.println (29%5%3*4−2);`

Solution:

Syntactically correct, value 2.

f) `System.out.println (29%5%3−4*2);`

Solution:

Syntactically correct, value −7.

g) `System.out.println (7%3 = 1);`

Solution:

Syntactically Incorrect.

The assignment operator `=` requires a variable on the left side. Danger of confusion with the equality operator `==`.

h) `System.out.println (1 / 0);`

Solution:

Syntactically correct, run-time error.

i) `1 / 0.0`

Solution:

Syntactically correct, value infinity.

j) `System.out.println (((10 − 2) * 4%)8));`

Solution:

Syntactically incorrect, `''''` expected.

k) `float x = 2.5;`
`float y = 3.4;`
`float z = x + y;`
`System.out.print ("The value is " + z );`

Solution:

Syntactically incorrect. By default in Java, a floating point number is represented as **double** with 64-bits. Thus, we can not put a **double** in **float** without writing the **'f'** suffix.

l) `int x = 34567456;`
`long y = 1234567;`
`long z = x + y;`
`System.out.println("The value is " + z );`
`y = 46547864784282;`
`z = x + y;`
`System.out.print ("The value is " + z );`

Solution:

Syntactically incorrect. By default in Java, a literal for integers is an **int** with 32-bits. If we want to store a number like 46547864784282 that exceeds the range of the **int**, we need to write the **'l'** suffix..

```
m) byte x = 12;
   byte y = 8;
   byte z = x + y;
   System.out.print ("The value is " + z );
```

Solution:

Syntactically incorrect. On the hardware level the '+' operation is done on **int** values. Thus, we need to typecast it into **byte** as well.

```
n) byte b = (byte) 400;
   System.out.println(b);
```

Solution:

```
-112
400 seen as an int (32 bits) by Java that is equal to:
00000.....0110010000
Since we are casting to byte (8 bits), the first 8 bits are truncated to become:
10010000 which is equal to:
-2^7 + 16 = -128 + 16 = -112
```

```
o) byte b = (byte) 128;
   System.out.println(b);
```

Solution:

```
-128
```

Exercise 1-3 Time
To be discussed in tutorial

Write an algorithm that reads the amount of time in seconds and then displays the equivalent hours, minutes and remaining seconds.

- One hour corresponds to 60 minutes.
- One minute corresponds to 60 seconds.

Solution:

```
import java.util.Scanner;
public class Time
{
    public static void main(String [] args) {
        Scanner sc = new Scanner(System.in);
        int seconds, minutes, hours;
        System.out.println("Enter number of seconds");
        seconds = sc.nextInt();
        hours = seconds / 3600;
        seconds = seconds % 3600;

        minutes = seconds / 60;
        seconds = seconds % 60;
```

```

        System.out.println("Hours:_" +hours);
        System.out.println("Minutes:_" +minutes);
        System.out.println("Seconds:_" +seconds);
    }
}

```

Exercise 1-4 Number Precision, Exactness
To be solved in Labs

Given the snippet of code below, what will be the values of d3 and d4 that will be printed?

```

double d1 = 1.03;
double d2 = 0.42;
double d3 = d1 - d2;
System.out.println(d3);

double d4 = 0.1;
float f    = (float) d4;
d4         = 1 - f;
System.out.println(d4);

```

Solution:

d3 will equal 0.6100000000000001, and d4 will equal 0.8999999761581421.

Exercise 1-5 Cook in a Hurry
To be solved in Lab

You want to cook some pasta, but you are short of time. To be sure you can finish cooking before your next appointment, you need to know how long it takes for the water to boil.

At its highest setting, your stove needs two minutes per liter ($1l = \frac{1}{1,000}m^3$) to reach the boiling point. You use a cylindrical pot.

Write a program that, given the diameter of the pot and the height of the water in it, calculates the the time needed for the water to boil.

Hint. The Volume of a cylinder is $\pi \times r^2 \times h$.

Solution:

```

public class CookSomePasta {
    public static void main (String[] args) {
        final double PI = 3.1415;
        double diameter = 7.0;
        double height   = 6.0;
        double radius    = diameter / 2;

        // calculate the volume
        double volume    = PI * radius * radius * height;

        // Assume that diameter & height are given in centimeters
        double liters    = volume / 1000;

        // The stove needs two minutes per liter
        double time      = liters * 2;
    }
}

```

```

        System.out.print("Stove_needs_" + time + "_minutes.");
    }
}

import java.util.*;
public class CookSomePasta {
    public static void main (String[] args) {
        final double PI = 3.1415;
        Scanner sc      = new Scanner(System.in);

        System.out.print("Enter_diameter:");
        double diameter = sc.nextDouble();

        System.out.print("Enter_height:");
        double height   = sc.nextDouble();

        double radius   = diameter / 2;

        // calculate the volume
        double volume    = PI * radius * radius * height;

        // Assume that diameter & height are given in centimeters
        double liters    = volume / 1000;

        // The stove needs two minutes per liter
        double time      = liters * 2;

        System.out.print("Stove_needs_" + time + "_minutes.");
    }
}

```

Exercise 1-6 Octal Conversion
To be solved in Lab

Assume that you have an octal number (base 8) and you would like to convert it into its equivalent decimal number (base 10) and binary number (base 2). For example: $(27)_8$ is equal to $(23)_{10}$ in decimal and $(10111)_2$ in binary.

Write a sequential Java program to convert a two digit octal number into its equivalent binary and decimal numbers.

Solution:

```

import java.util.*;
public class Octal
{
    public static void main (String[] args) {
        int oct0, oct1;
        int q0, q1, q2, r0, r1, r2;
        Scanner sc = new Scanner(System.in);
        System.out.print("Please_enter_the_octal_number_");
        int octal = sc.nextInt();

        //extracting the digits
        oct0 = octal%10;
        oct1 = octal/10;

        // Calculate the decimal Value
        double dec = oct0 * 1 + oct1 * 8;
        int result = (int) dec;
    }
}

```

```

        System.out.println(octal + "_in_octal_=" + result + "_in_decimal.");

        //converting the first digit base 8 into 3 bits
        q0 = oct1 / 2;
        r0 = oct1 % 2;
        q1 = q0 / 2;
        r1 = q0 % 2;
        q2 = q1 / 2;
        r2 = q1 % 2;
        System.out.print(octal + "_in_octal_=" + r2 + "_" + r1 + "_" + r0 + "_");

        //converting the second digit base 8 into 3 bits
        q0 = oct0 / 2;
        r0 = oct0 % 2;
        q1 = q0 / 2;
        r1 = q0 % 2;
        q2 = q1 / 2;
        r2 = q1 % 2;
        System.out.println(r2 + "_" + r1 + "_" + r0 + "_in_binary.");
    }
}

```

Exercise 1-7 Supermarket Change

Write a Java program CountChange to count change. Given the number of quarter, dimes, nickles, and pennies the program should output the total as a single value in dollars and pennies.

Hint:

- One dollar corresponds to 100 pennies.
- One quarter corresponds to 25 pennies.
- One dime corresponds to 10 pennies.
- One nickle corresponds to 5 pennies.

For example if we have: 3 quarters, 2 dimes, 1 nickle and 6 pennies, then the total is 1.06 dollars.

Solution:

Initializing variables by value:

```

public class CountChange
{
    public static void main(String [] args) {
        int quarters , dimes , nickles , pennies;
        quarters = 3;
        dimes    = 2;
        nickles  = 1;
        pennies  = 6;
        double total = quarters * 25 + dimes * 10 + nickles * 5 + pennies;
        System.out.println("Total_ is:_" + total / 100);
    }
}

```

Initializing variables by input:

```

import java.util.Scanner;
public class CountChange

```

```

{
    public static void main(String [] args) {
        Scanner sc = new Scanner(System.in);
        int quarters, dimes, nickles, pennies;
        System.out.println("Enter_number_of_quarters");
        quarters = sc.nextInt();
        System.out.println("Enter_number_of_dimes");
        dimes = sc.nextInt();
        System.out.println("Enter_number_of_nickles");
        nickles = sc.nextInt();
        System.out.println("Enter_number_of_pennies");
        pennies = sc.nextInt();
        double total = quarters * 25 + dimes * 10 + nickles * 5 + pennies;
        System.out.println("Total_is:_$" + total/100);
    }
}

```

Exercise 1-8 Electrical Resistance

The equivalent resistance of resistors connected in series is calculated by adding the resistances of the individual resistors.

The formula for resistors connected in parallel is a little more complex. Given two resistors with resistances R1 and R2 connected in parallel the equivalent resistance is given by the inverse of the sum of the inverses; e.g.

$$\frac{1}{R_{eq}} = \frac{1}{R1} + \frac{1}{R2}.$$

Write a program that given 3 resistances outputs the equivalent resistance when they are connected in series and when they are connected in parallel.

Solution:

```

public class Resistance
{
    public static void main (String [] args)
    {
        float r1 = 8, r2 = 8, r3 = 4;
        float resS = r1 + r2 + r3;
        float resPInv = 1/r1 + 1/r2 + 1/r3;
        float resP = 1 / resPInv;

        System.out.println("Series_resistance:" + resS);
        System.out.println("Parallel_resistance:" + resP);
    }
}

```

Exercise 1-9 Temperature Conversion

Assume that you have a Celsius scale temperature of 100 degrees and you wish to convert it into degrees on the Fahrenheit scale and the Kelvin scale. This conversion is done using the following formulas.

$$T_f = \frac{9}{5} \times T_c + 32, \text{ and}$$

$$T_k = T_c + 273.$$

Where T_f is temperature in degrees Fahrenheit, T_c is temperature in degrees Celsius, and T_k is temperature in degrees Kelvin.

Write a Java program that takes as an input a degree in Celsius scale and converts it into both Fahrenheit and Kelvin scales.

Solution:

```
import java.util.*;
public class TemperatureCalculator {
    public static void main (String [] args) {
        Scanner sc = new Scanner(System.in);
        double celsius , fahrenheit , kelvin;

        System.out.print("Please insert the temperature in degrees Celsius ");
        celsius = sc.nextDouble();
        fahrenheit = (9.0/5 * celsius) + 32;
        kelvin = celsius + 237 ;

        System.out.println("Temperature in degrees Fahrenheit is " + fahrenheit);
        System.out.println("Temperature in degrees Kelvin is " + kelvin);
    }
}
```